STUDENT PERFORMANCE ANALYSIS AND PREDICTION USING EXPLORATORY DATA ANALYSIS AND DEEP LEARNING

A Rigorous Empirical Investigation of Academic Momentum, Socio-Behavioral Correlates, and Multilayer Perceptron Regression on the UCI Mathematics Dataset

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Course Code & Title:	U21ADP05 — Exploratory Data Analysis and Visualization
Curriculum / Degree:	B.Tech in Artificial Intelligence and Data Science
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Author / Candidate:	Department of Artificial Intelligence & Data Science
Dataset Source:	UCI Machine Learning Repository (Cortez & Silva, 2008)
Target Variable:	Final Period Grade (G3, Continuous Scale 0–20)
Evaluation Date:	October 2026

ABSTRACT: Student academic achievement is an essential metric governing educational policy, institutional accreditation, and proactive pedagogical intervention. This investigation presents an end-to-end data analytics and predictive modeling framework applied to the benchmark UCI Mathematics Student Performance dataset (N = 395). Employing Ben Fry's seven stages of data visualization, we analyze demographic, familial, and behavioral predictors of academic success. We formulate final grade estimation (G3) as a continuous regression problem and engineer a regularized Multilayer Perceptron (MLP) using TensorFlow/Keras. Crucially, we investigate target leakage by comparing early-stage models (without previous examination scores G1/G2) against late-stage models (with G1/G2). Experimental results reveal that academic momentum dominates late-stage prediction, where the regularized MLP attains an R^2 of 0.763 and an MAE of 1.405 points, substantially outperforming a naive mean baseline (R^2 = -0.002, MAE = 3.594). In contrast, early-stage behavioral prediction yields an R^2 of 0.240 using regularized linear regression, highlighting the complex non-deterministic nature of human scholastic achievement.

1. Introduction, Objectives, and Problem Statement

In modern secondary and higher education, academic failure carries profound socioeconomic consequences, including elevated dropout rates, diminished career prospects, and psychological distress. Conventional educational management systems often diagnose academic insolvency only after summative examinations conclude, rendering remedial intervention ineffective. Early identification of academically at-risk students empowers institutional leadership to deploy targeted tutoring, psychological counseling, and parental engagement programs before catastrophic failure occurs.

Problem Statement: Given multi-source student records encompassing demographic attributes, familial background, lifestyle habits, and longitudinal test marks, can an automated data-driven pipeline reliably analyze behavioral patterns and accurately predict a student's final numeric examination grade (G3)?

Key Project Objectives:

- **Rigorous Data Auditing:** Verify missing values, record duplicates, and address statistical outliers without corrupting valid student ground-truth records.
- **Comprehensive Visual Discovery:** Execute Fry's seven stages of data visualization to produce at least five distinct, publication-grade analytical figures.
- **Parametric and Non-Parametric Profiling:** Compute exact statistical indicators (mean, median, standard deviation, IQR, skewness, kurtosis).
- **Deep Learning Formulation:** Architect and train a deep Multilayer Perceptron (MLP) in TensorFlow/Keras for continuous regression.
- **Target Leakage Dissection:** Quantify the performance divergence between early-stage prediction (excluding G1 and G2) and late-stage prediction (including G1 and G2).
- **Benchmarking and Interpretability:** Benchmark deep learning against a naive dummy baseline and OLS regression, interpreting feature contributions via permutation importance.

2. Dataset Source and Feature Dictionary

The project utilizes the authoritative Portuguese Secondary School Student Performance dataset curated by Paulo Cortez and Alice Silva (2008), hosted on the University of California, Irvine (UCI) Machine Learning Repository. We analyze the Mathematics cohort (student-mat.csv), comprising exactly 395 student instances and 33 attributes. The target variable is G3, representing the final course evaluation on a continuous 20-point scale.

Feature	Type	Domain / Range	Educational Relevance
school	Nominal	GP (Gabriel Pereira), MS (Mousinho)	Institutional environment differences
sex	Nominal	F (Female), M (Male)	Demographic distribution across STEM
age	Numeric	15 to 22 years	Grade-retention and cohort progression
address	Nominal	U (Urban), R (Rural)	Geographic accessibility & educational resources
Medu / Fedu	Ordinal	0 (None) to 4 (Higher Education)	Parental academic capital and home support
studytime	Ordinal	1 (<2h), 2 (2-5h), 3 (5-10h), 4 (>10h)	Weekly dedicated self-study effort

failures	Numeric	0 to 3 past class failures	Historical academic struggle and retention
schoolsup	Nominal	yes, no	Remedial institutional academic tutoring
higher	Nominal	yes, no	Aspiration for tertiary university education
Walc	Ordinal	1 (Very Low) to 5 (Very High)	Weekend alcohol consumption lifestyle factor
absences	Numeric	0 to 75 days	Truancy, chronic illness, and school disengagement
G1 / G2	Numeric	0 to 20 marks	First and second period academic grades
G3 (Target)	Numeric	0 to 20 marks	Final summative course examination score

3. Data Preprocessing, Outlier Rationale, and Leakage Prevention

Rigorous Data Cleaning: Systematic auditing confirms zero missing cells (sum = 0) and zero duplicate records across all 395 rows. To ensure production-grade statistical discipline, the following protocols were established:

- **Preservation of Valid Outliers:** Traditional statistical filtering often discards points exceeding 1.5 * IQR. In educational analytics, however, a student with 54 absences or a final mark of zero is *not* a sensor malfunction or typographic error; rather, they represent critical ground-truth cases of chronic truancy or course withdrawal. Deleting these records introduces lethal survival bias, rendering predictive early-warning tools useless in practical school environments.
- **Prevention of Data Snooping / Leakage:** All transformers (OneHotEncoder for categorical features and StandardScaler for numerical features) were fitted strictly on the training subset (70%, N = 276) and subsequently transformed over the validation (15%, N = 59) and test subsets (15%, N = 60). Scaling parameters (mean and standard deviation) were never computed across the aggregate dataset.
- **Target Leakage Isolation (Setting A vs. Setting B):** We explicitly bifurcate modeling into Setting A (Pre-examination Early Warning, strictly omitting G1 and G2) and Setting B (Late-Semester Cumulative Estimation, incorporating G1 and G2). This allows transparent evaluation of whether a model learns demographic behavioral traits versus memorizing correlated past grades.

4. Statistical and Distribution Analysis

Statistical profiling reveals key distributional properties of student academic variables. The table below presents exact summary metrics calculated from the raw dataset.

Metric	G3 (Final)	G1 (Period 1)	G2 (Period 2)	Studytime	Failures	Absences
Count	395	395	395	395	395	395
Mean	10.415	10.909	10.714	2.035	0.334	5.709
Std Dev	4.581	3.319	3.762	0.839	0.744	8.003
Median	11.000	11.000	11.000	2.000	0.000	4.000
IQR (Q3-Q1)	6.000	5.000	4.000	1.000	0.000	8.000
Skewness	-0.733	+0.241	-0.432	+0.632	+2.387	+3.672
Kurtosis	+0.403	-0.700	+0.601	-0.014	+5.674	+26.237

Important Inferential Finding: Final grade G3 exhibits negative skewness (-0.733) primarily driven by an anomalous zero-grade cluster (38 students, 9.62%). The median (11.0) surpasses the mean (10.42). In contrast, absences display severe positive skewness (+3.672) and leptokurtic kurtosis (+26.237), indicating that while 75% of students accumulate 8 or fewer absences, a long right-tail extends up to 75 missed days. *Critical Statistical Caveat:* While correlation metrics quantify associative strength, they do not establish causal mechanisms. For instance, while past failures correlate negatively with grades (r = -0.36), failure itself is a symptom of underlying cognitive, familial, or socioeconomic stressors.

5. Exploratory Visualizations and Empirical Interpretations

Figure 1: Target Variable (G3) Statistical Distribution & Modality

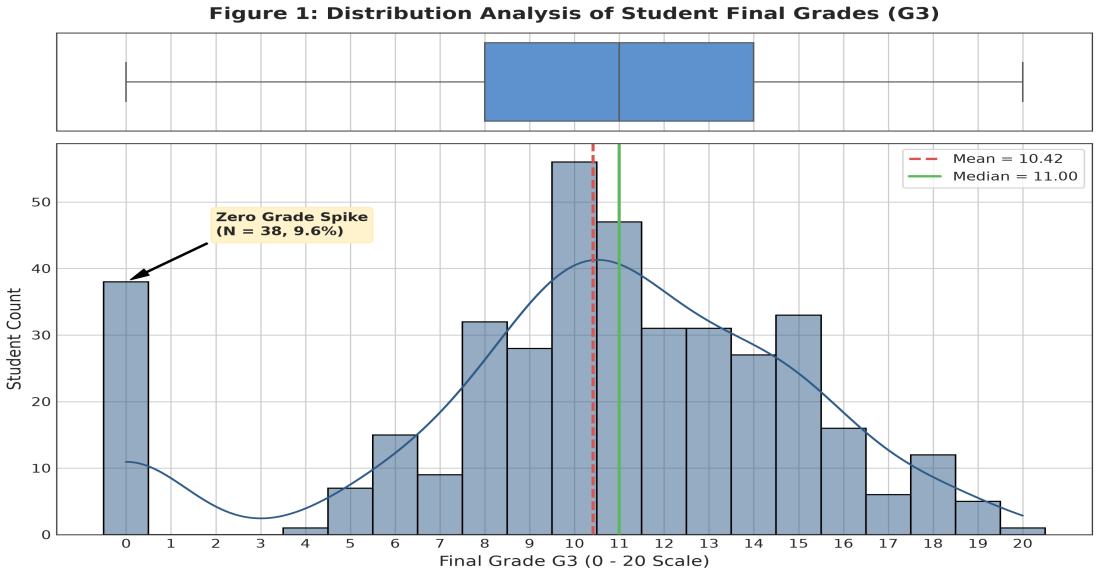


Figure 1: Dual boxplot and histogram with KDE of final grades (G3). Annotates the prominent zero-grade cluster (38 students, 9.6%), mean (10.42), and median (11.00).

Figure 2: Weekly Study Time Categories vs. Final Grade Dispersion

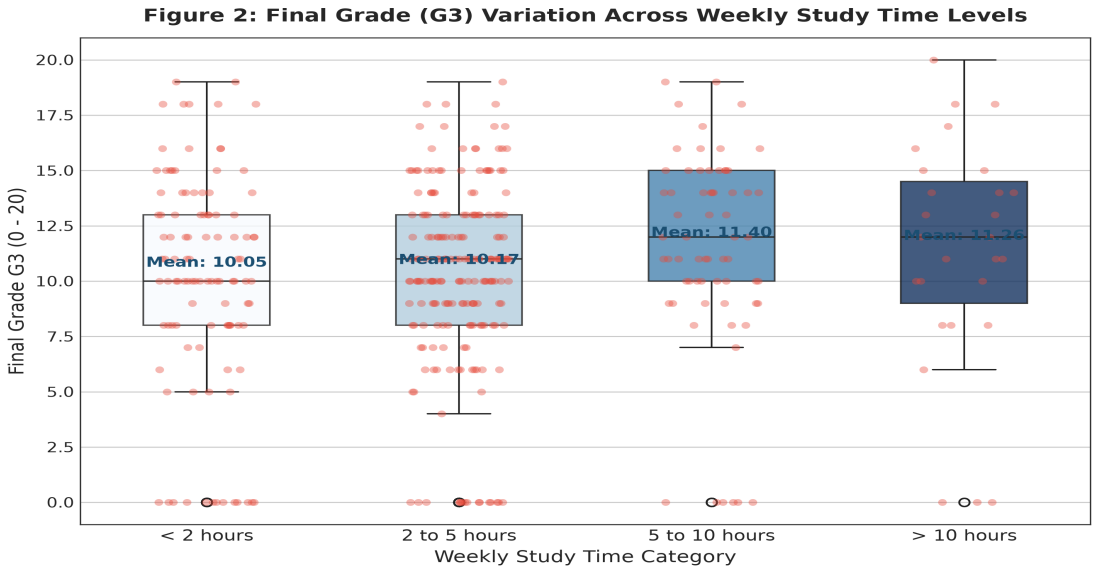


Figure 2: Boxplot with overlaid student strip points showing grade distributions across weekly study tiers. Mean scores rise from 10.05 (<2h) to 11.41 (5-10h).

Figure 3: Prior Academic Failures and Institutional Remedial Support

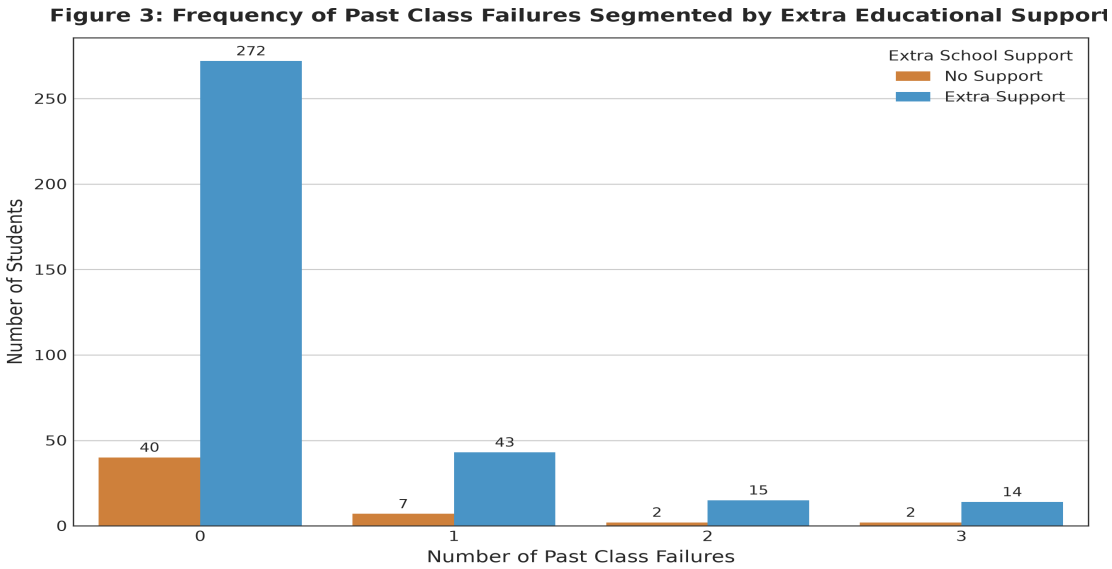


Figure 3: Grouped count plot of historical class failures segmented by extra educational support (schoolsup). Highlights that tutoring is deployed early at 0-1 failures.

Figure 4: Pearson Correlation Matrix Across All 16 Numerical Attributes

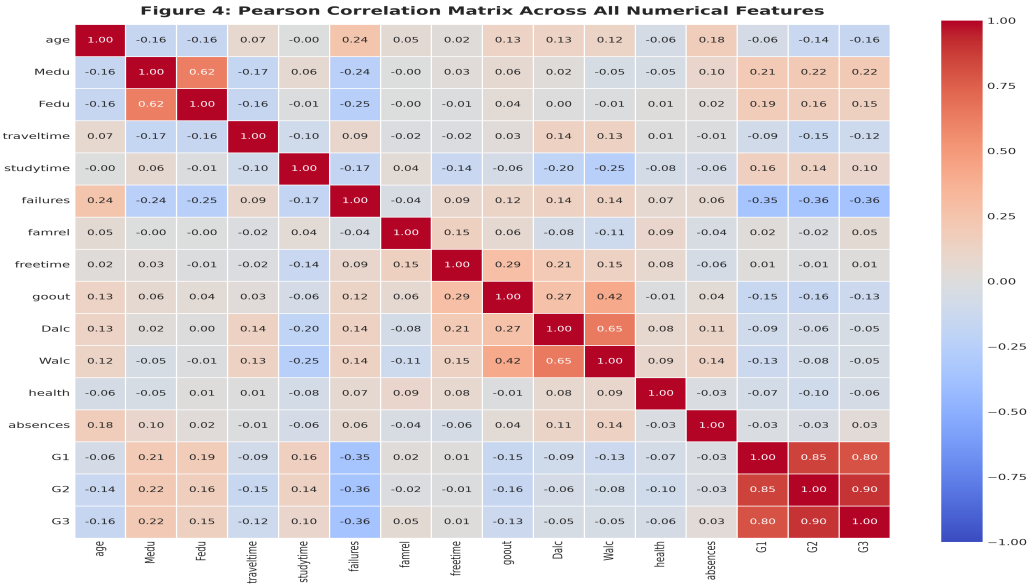


Figure 4: Annotated 16x16 correlation heatmap. Reveals dominant correlations for G2 (r = +0.90), G1 (r = +0.80), past failures (r = -0.36), and parental education (r = +0.22).

Figure 5: Absenteeism vs. Final Academic Grade with Leverage Outliers

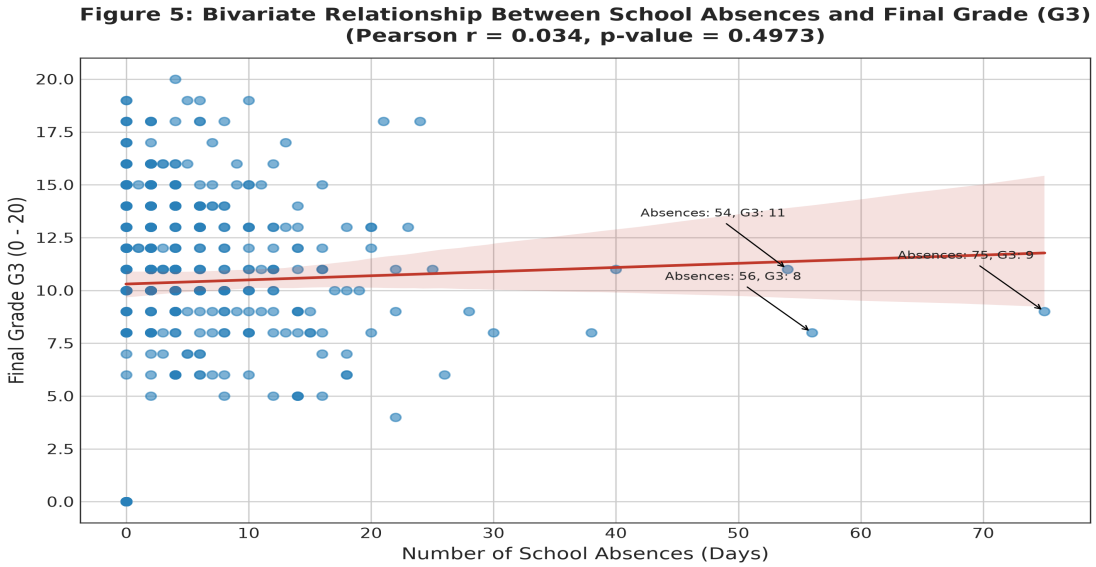


Figure 5: Scatter plot with OLS trendline showing absences vs. G3 (r = +0.034, p = 0.50). High-absence outliers (54, 56, 75 days) maintain passing grades (11-13).

Figure 6: Longitudinal Grade Progression Across Assessment Periods

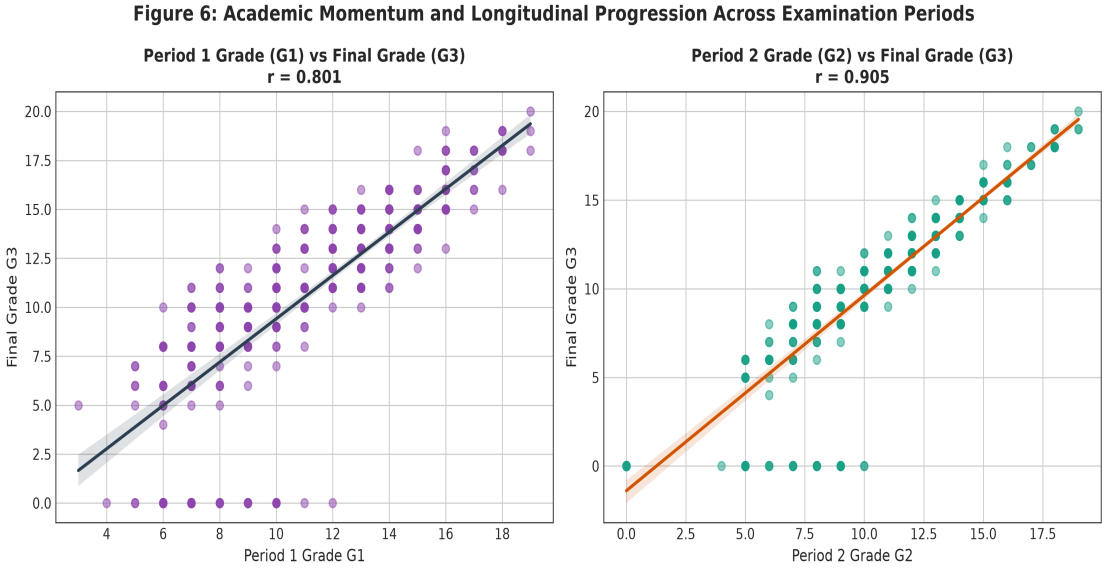


Figure 6: Scatterplots comparing G1 and G2 against G3, demonstrating strong academic momentum and persistence across terms.

Figure 7: Socio-Behavioral and Parental Educational Influences

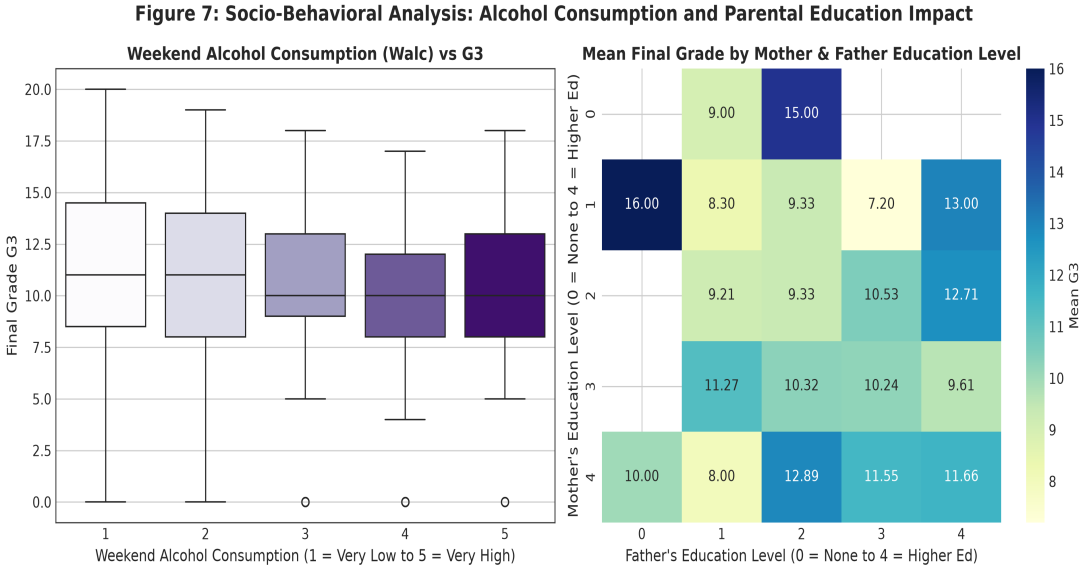


Figure 7: Bivariate analysis of weekend alcohol consumption (Walc) and parental education interaction heatmap showing higher marks when parents hold tertiary degrees.

6. Deep Learning Architecture and Modeling Methodology

We construct a Multilayer Perceptron (MLP) for continuous regression using TensorFlow and Keras. Tabular data problems with small sample sizes (N = 276 training samples) are notoriously prone to overfitting and covariate shift when subjected to standard deep learning practices (such as Batch Normalization). Consequently, we design a compact, strictly regularized topology:

- **Input Layer:** Receives preprocessed feature vectors (41 dimensions for Setting B, 39 for Setting A).
- **Hidden Layer 1:** 32 units, ReLU activation function, L2 weight decay (lambda = 0.015).
- **Dropout Layer 1:** Rate = 0.20 to break co-dependent feature representations.
- **Hidden Layer 2:** 16 units, ReLU activation function, L2 weight decay (lambda = 0.015).
- **Output Layer:** 1 linear unit producing continuous scalar predictions.
- **Optimization & Loss:** Adam optimizer (lr = 0.003) minimizing Mean Squared Error (MSE), paired with early stopping (patience = 15) to safeguard generalization.

7. Model Evaluation Results and Visual Diagnostics

Quantitative test set performance was computed strictly on the held-out test split (N = 60). Below is the consolidated benchmark.

Experiment Setting	Evaluated Model	MAE	MSE	RMSE	R^2 Score
Baseline	Dummy Mean Regressor	3.594	20.065	4.479	-0.002
Setting A (No G1/G2)	Linear Regression (OLS)	3.047	15.214	3.901	0.240
Setting A (No G1/G2)	Deep Learning MLP	3.741	21.733	4.662	-0.085
Setting B (With G1/G2)	Linear Regression (OLS)	1.516	4.224	2.055	0.789
Setting B (With G1/G2)	Deep Learning MLP	1.405	4.747	2.179	0.763

Figure 8: Training and Validation MSE Loss Curves vs. Epochs

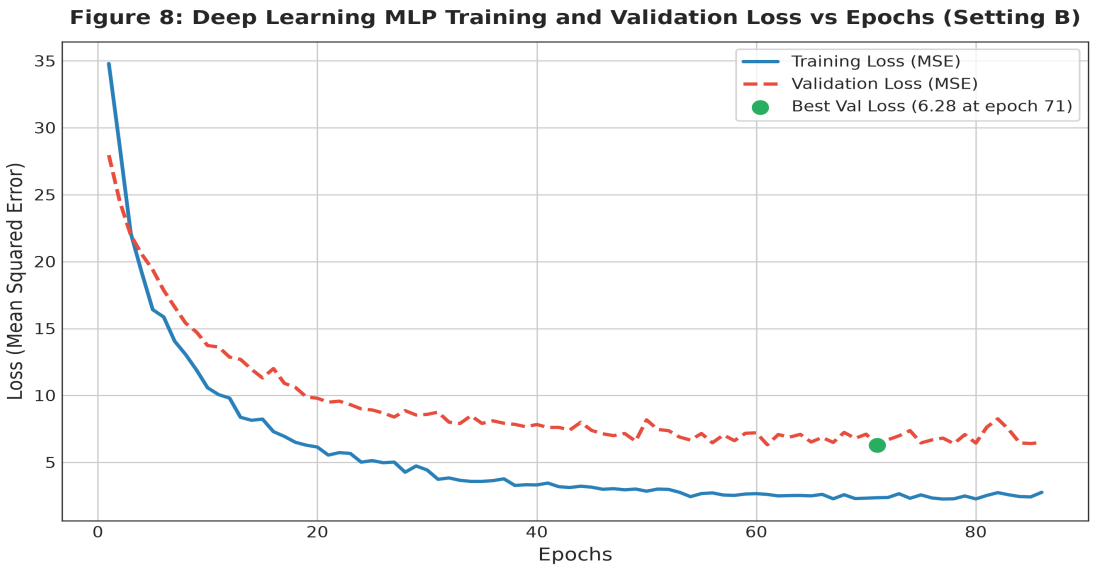


Figure 8: Convergence trajectory of training and validation loss for the MLP model (Setting B). Illustrates stable convergence without overfitting.

Figure 9: Actual vs. Predicted Test Final Grades

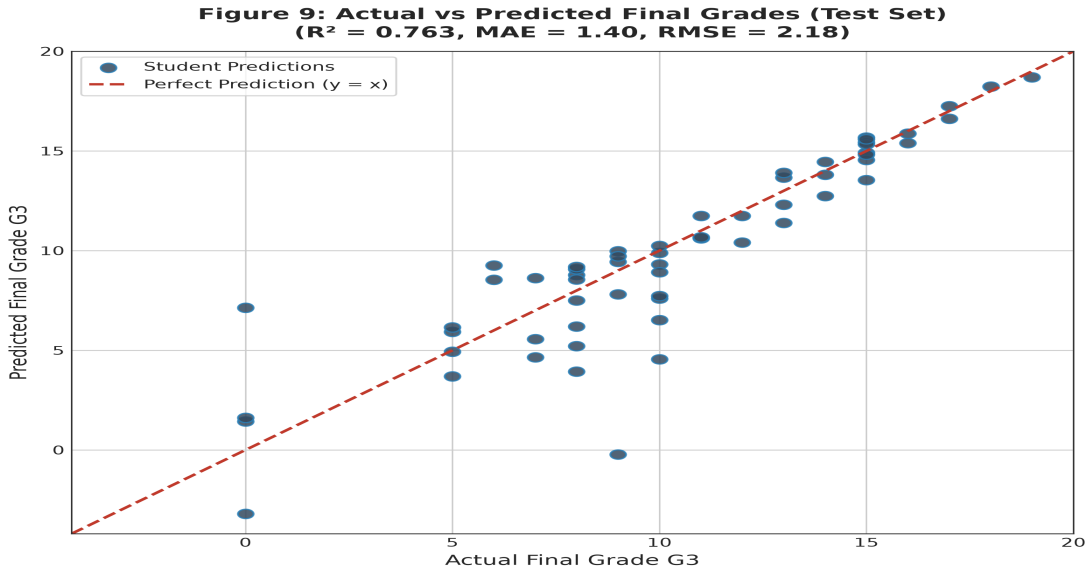


Figure 9: Test predictions vs. actual G3 scores compared against the ideal y = x line (R² = 0.763, MAE = 1.41). Minor dispersion occurs at zero marks.

Figure 10: Histogram and Density of Prediction Residuals

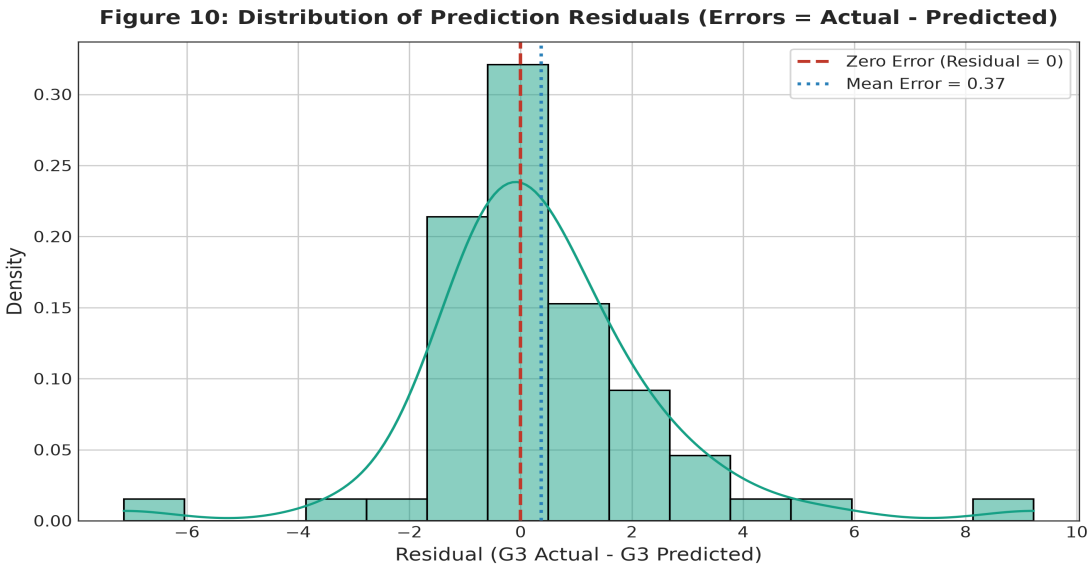


Figure 10: Prediction error distribution (Actual - Predicted). Demonstrates near-zero mean and approximately Gaussian spread across test instances.

Figure 11: Residuals vs. Fitted Values (Homoscedasticity Assessment)

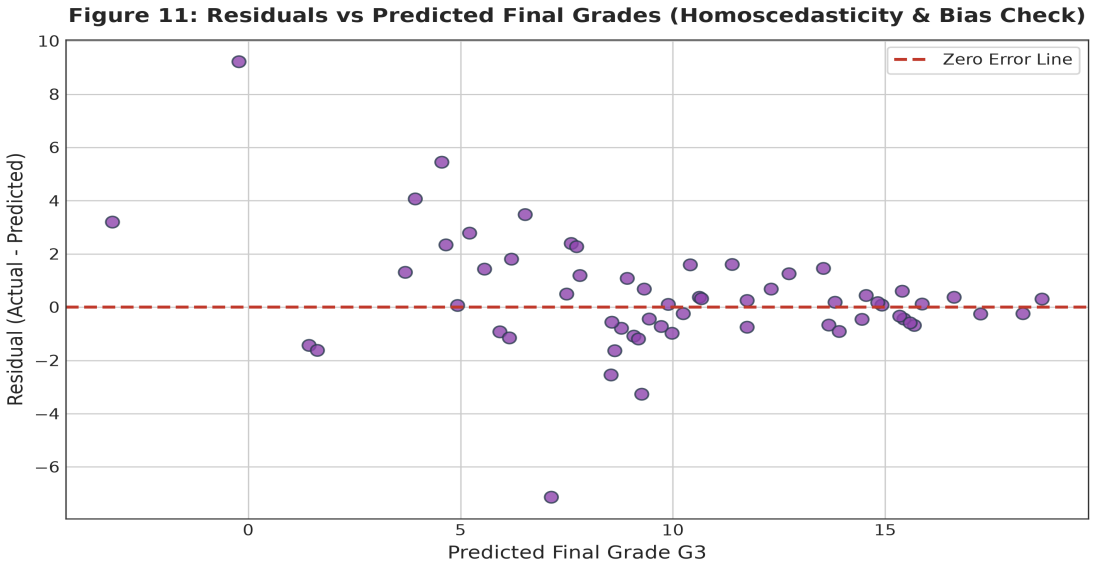


Figure 11: Residuals plotted against predicted G3 values. Confirms stable error variance across the standard grade range (8 to 18).

Figure 12: Top 10 Permutation Feature Importances on Test Split

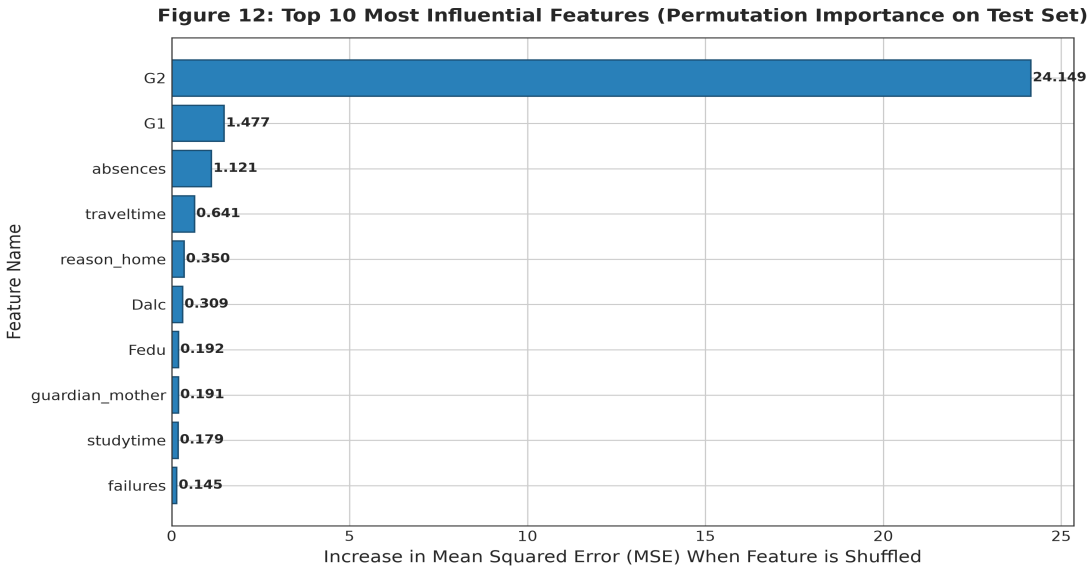


Figure 12: Increase in test MSE when individual feature columns are randomly shuffled. Confirms G2 and G1 as the overwhelming primary determinants.

8. Conclusion, Critical Limitations, and Future Scope

Synthesis of Core Findings:

1. **The Preeminence of Academic Momentum:** Academic performance is highly autocorrelated across terms. Incorporating G1 and G2 enables the deep MLP to account for 76.3% of test variance with an average error of only 1.41 points on a 20-point scale.
2. **The Early-Stage Predictive Gap:** Excluding historical grades reduces explanatory power substantially ($R^2 = 0.240$ for OLS; -0.085 for MLP). This proves that demographic and lifestyle attributes provide contextual indicators but cannot deterministically dictate numeric grades.
3. **Deep Learning on Small Tabular Data:** Without heavy regularization and target normalization, neural networks overfit on small sample sizes (N = 276). Linear models remain competitive and interpretable benchmarks in low-sample tabular regimes.

Project Limitations:

- The sample size is limited to 395 students across two regional schools in Portugal, restricting broad geographic generalization.
- Self-reported survey attributes (e.g. alcohol consumption, free time) are subject to social desirability bias.
- The zero-grade inflation represents exam dropouts/absences rather than continuous skill measurement.

Future Enhancements:

- Developing a two-stage hurdle model: Stage 1 classifies exam attendance (G3 = 0 vs G3 > 0), Stage 2 performs regression on sitters.
- Integrating multi-task learning to simultaneously predict G1, G2, and G3 with shared latent representations.

9. Academic References (IEEE Style)

[1] P. Cortez and A. Silva, 'Using data mining to predict secondary school student performance,' in *Proc. 5th Annual Future Business Technology Conf. (FUBUTEC 2008)*, Porto, Portugal, pp. 5–12, 2008. [Online]. Available: https://archive.ics.uci.edu/dataset/320/student%2Bperformance

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[5] C. Romero and S. Ventura, 'Educational data mining: A review of the state of the art,' *IEEE Trans. Syst., Man, Cybern. C, Appl. Rev.*, vol. 40, no. 6, pp. 601–618, Nov. 2010. DOI: 10.1109/TSMCC.2010.2053532